

Problem E. Is it a tree

Time Limit	500 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

You are given an unweighted, undirected graph. Write a program to check if it's a tree topology.

Input

The first line of the input file contains two integers N and M – number of nodes and number of edges in the graph ($0 < N \leq 10000$, $0 \leq M \leq 20000$). Next M lines contain M edges of that graph – Each line contains a pair (u, v) means there is an edge between node u and node v ($1 \leq u, v \leq N$).

Output

Print *YES* if the given graph is a tree, otherwise print *NO*.

Example

Input	Output
3 2 1 2 2 3	YES